



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025-26
MONDAY TEST

CLASS: X
SUBJECT: COMPUTER APPLICATIONS

TOTAL MARKS: 40
DATE: 16.05.25

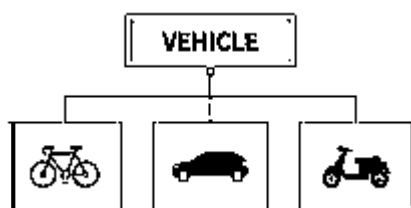
Attempt all questions. The intended marks for questions are given in brackets [].
This paper consists of three printed pages.

Question 1

[10x1=10]

Choose the correct answers to the questions from the given options

(i)



Identify the feature of Java depicted by the above picture:

(a) Encapsulation (b) Data Abstraction (c) Inheritance (d) Class And Object

(ii) Which function is not overloaded in the Math class?

(a) Math.max() (b) Math.pow() (c) Math.min() (d) Math.abs()

(iii) Choose the correct output:

`System.out.println(Math.pow((Math.max(Math.ceil(-8.5),Math.floor(-8.8))),2));`

(a) 81.0 (b) 81 (c) 72.0 (d) 64.0

(iv) What is the return type of a method that does not return any value?

(a) null (b) void (c) empty (d) zero

(v) Assertion(A): Method signatures in Java include the method name and parameter list.

Reason(R): Return type is not a part of the method signature in Java.

(a) Both Assertion (A) and Reason (R) are true and (R) is a correct explanation of (A)
(b) Both Assertion (A) and Reason (R) are true and (R) is not a correct explanation of (A)
(c) Assertion (A) is true and Reason (R) is false
(d) Assertion (A) is false and Reason(R) is true

(vi) What will happen if a method that is supposed to return a value doesn't have the return statement?

(a) Compilation error (b) Runtime error (c) No output (d) The method runs as void

(vii) Choose the correct output:

```
int i=1;
```

```
for ( ; i<= 3; i++);
```

```
System.out.print(i + " ");
```

(a) 123

(b) 1

(c) 1234

(d) 4

(viii) Choose the correct output if n=4:

```
String s = "3.14";
```

```
float f = Float.parseFloat(s);
```

```
System.out.println(f + 1);
```

(a) 4.14

(b) Error

(c) 3.141

(d) 3.14+1

(ix) How many times will the outer loop execute?

```
for (int i = 1; i <= 5; i++) {
```

```
    for (int j = i; j <= 5; j++) {
```

```
        System.out.print("*"); } 
```

```
    System.out.println();    }
```

(a) 5

(b) 15

(c) 10

(d) 25

(x) Assertion(A): Method overloading in Java allows multiple methods with the same name but different parameters.

Reason(R): Java supports method overloading through inheritance only.

(a) Both Assertion (A) and Reason (R) are true and (R) is a correct explanation of (A)

(b) Both Assertion (A) and Reason (R) are true and (R) is not a correct explanation of (A)

(c) Assertion (A) is true and Reason (R) is false

(d) Assertion (A) is false and Reason(R) is true

Question 2

i. Write the Java expression for

[2]

$$\frac{a^x + b^y}{\sqrt[3]{x} + \sqrt[3]{y}}$$

ii. Identify the errors and rewrite the code.

[4]

```
class A {
    int squarerooot (int sq) {
        return Math.sqrt(sq); }

    static void add( int a, int b) {
        return a+b; }

    public static void main ( ) {
        A obj =new A( );
        double s= squarerooot(9);
        add(3,5); } }
```

iii. Write a statement to convert a String object to double data type.

[2]

iv. Write a method that accepts a character and checks if it is a whitespace or not. [2]

Question 3 [10]

Riverside transport company charges for the parcels of its customers as per the following specifications given below:

Class Name : Riverside

Member variables :

String name : to store the name of the customer

double w : to store the weight of the parcel in kg

double charge : to store the charge of the parcel

Member functions :

void accept() : to accept the name of the customer, weight of the parcel from the user

void calculate() :	<table><tr><th>Weight in kg</th><th>Charge per kg</th></tr><tr><td>Upto 10kgs</td><td>₹ 25 per kg</td></tr><tr><td>Next 20kgs</td><td>₹ 20 per kg</td></tr><tr><td>Above 30kgs</td><td>₹ 10 per kg</td></tr></table>	Weight in kg	Charge per kg	Upto 10kgs	₹ 25 per kg	Next 20kgs	₹ 20 per kg	Above 30kgs	₹ 10 per kg
Weight in kg	Charge per kg								
Upto 10kgs	₹ 25 per kg								
Next 20kgs	₹ 20 per kg								
Above 30kgs	₹ 10 per kg								

A surcharge of 5% is charged on the bill.

void print() : to print the name of the customer ,weight of the parcel, total bill inclusive of surcharge in a tabular form in the following format:

Name	Weight	Bill amount
------	--------	-------------

Define a class with the above mentioned specifications, create the main method, create an object and invoke the member methods.

Question 4 [10]

Design a class to overload a function fun() as follows:

void fun(): Method to display the followong pattern

```
1
2 4
3 5 7
6 8 10 12
9 11 13 15 17
```

void fun(int num): A Dudeney number is a positive integer that is a perfect cube such that the sum of its digits is equal to the cube root of the number. Write a program to input a number and check and print whether it is a Dudeney number or not.

Example:

Consider the number 512.

Sum of digits = 5 + 1 + 2 = 8 Cube root of 512 = 8

As Sum of digits = Cube root of Number hence 512 is a Dudeney number.